ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	3127771
ToLID	**xgChaSegu1**
Species	Charonia seguenzae
Class	Gastropoda
Order	Littorinimorpha

Genome Traits	Expected	Observed
Haploid size (bp)	1,016,308,473	4,994,662,406
Haploid Number	17 (source: ancestor)	35
Ploidy	2 (source: ancestor)	2
Sample Sex	Unknown	Unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for collapsed: 5.8.Q33

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Haploid size (bp) has >20% difference with Expected
. Observed Haploid Number is different from Expected

. QV value is less than 40 for collapsed
. Kmer completeness value is less than 90 for collapsed
. BUSCO single copy value is less than 90% for collapsed
. BUSCO duplicated value is more than 5% for collapsed
. More than 1000 gaps/Gbp for collapsed
. Not 90% of assembly in chromosomes for collapsed

Curator notes

. Interventions/Gb: 190
. Contamination notes: ""
. Other observations: "The assembly of Charonia seguenzae (xgChaSegu1) is based on 58.9X ONT data and 137.09X Arima Hi-C data generated as part of the European Reference Genome Atlas (ERGA, https://www.erga-biodiversity.eu/) via the Biodiversity Genomics Europe project (BGE, https://biodiversitygenomics.eu/). The assembly process included the following steps: initial ONT assembly generation with Hifiasm, removal of contaminant sequences using Context, removal of haplotypic duplications using purge_dups, and Hi-C-based scaffolding with YaHS. In total, 888 contigs were identified as contaminants (bacterial, archaeal, or viral), totaling 57.417 Mb (with the largest being 0.479 Mb). Additionally, 17814 regions totaling 414.712 Mb (with the largest being 0.353 Mb) were identified as haplotypic duplications and removed.

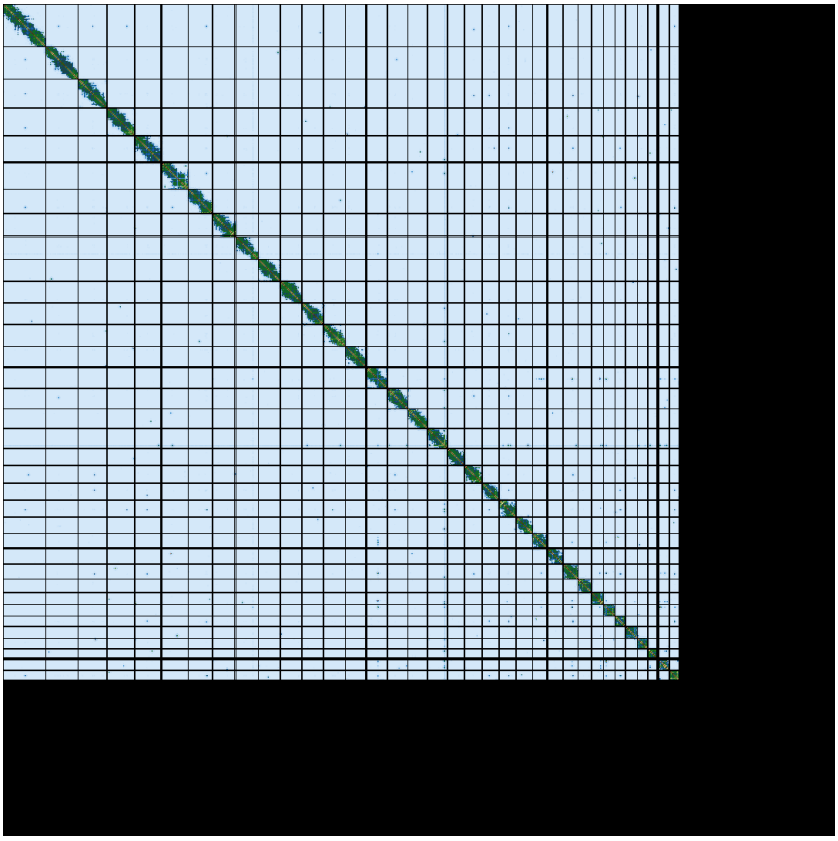
Mitochondrial genome was assembled using ptGAUL. Finally, the primary assembly was analyzed and manually improved using Pretext. Chromosome-scale scaffolds confirmed by Hi-C data were named in order of size. "

Quality metrics table

Metrics	Pre-curation collapsed	Curated collapsed
Total bp	5,002,866,701	4,994,662,406
GC %	43.82	43.82
Gaps/Gbp	4,413.07	4,473.58
Total gap bp	2,207,800	2,295,100
Scaffolds	17,889	14,252
Scaffold N50	79,325,356	112,930,338
Scaffold L50	21	18
Scaffold L90	2,372	2,029
Contigs	39,967	36,596
Contig N50	256,834	257,473
Contig L50	4,382	4,367
Contig L90	21,931	21,801
QV	33.432	33.434
Kmer compl.	86.7935	86.6938
BUSCO sing.	70.2%	76.1%
BUSCO dupl.	20.5%	14.6%
BUSCO frag.	6.0%	5.9%
BUSCO miss.	3.3%	3.3%

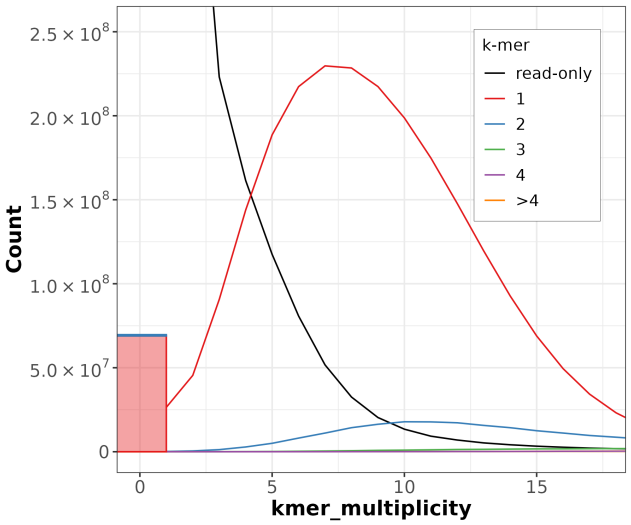
Warning! BUSCO versions or lineage datasets are not the same across results:
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: mollusca_odb12 (genomes:36, BUSCOs:4421)
BUSCO: 6.1.0 (euk_genome_min, miniprot) / Lineage: mollusca_odb12 (genomes:36, BUSCOs:4421)

HiC contact map of curated assembly

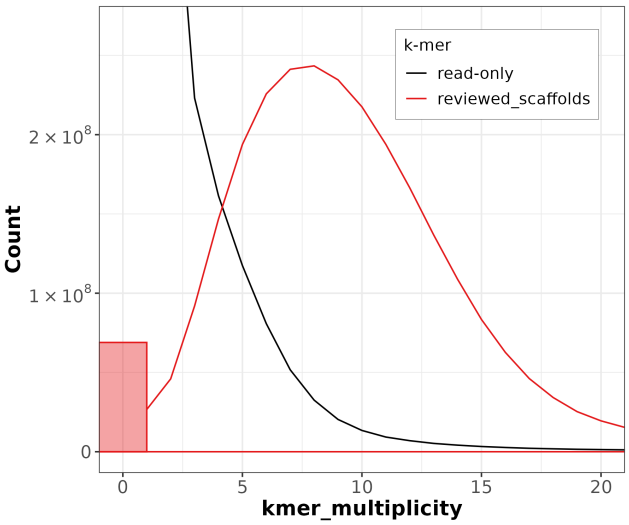


collapsed [LINK]

K-mer spectra of curated assembly

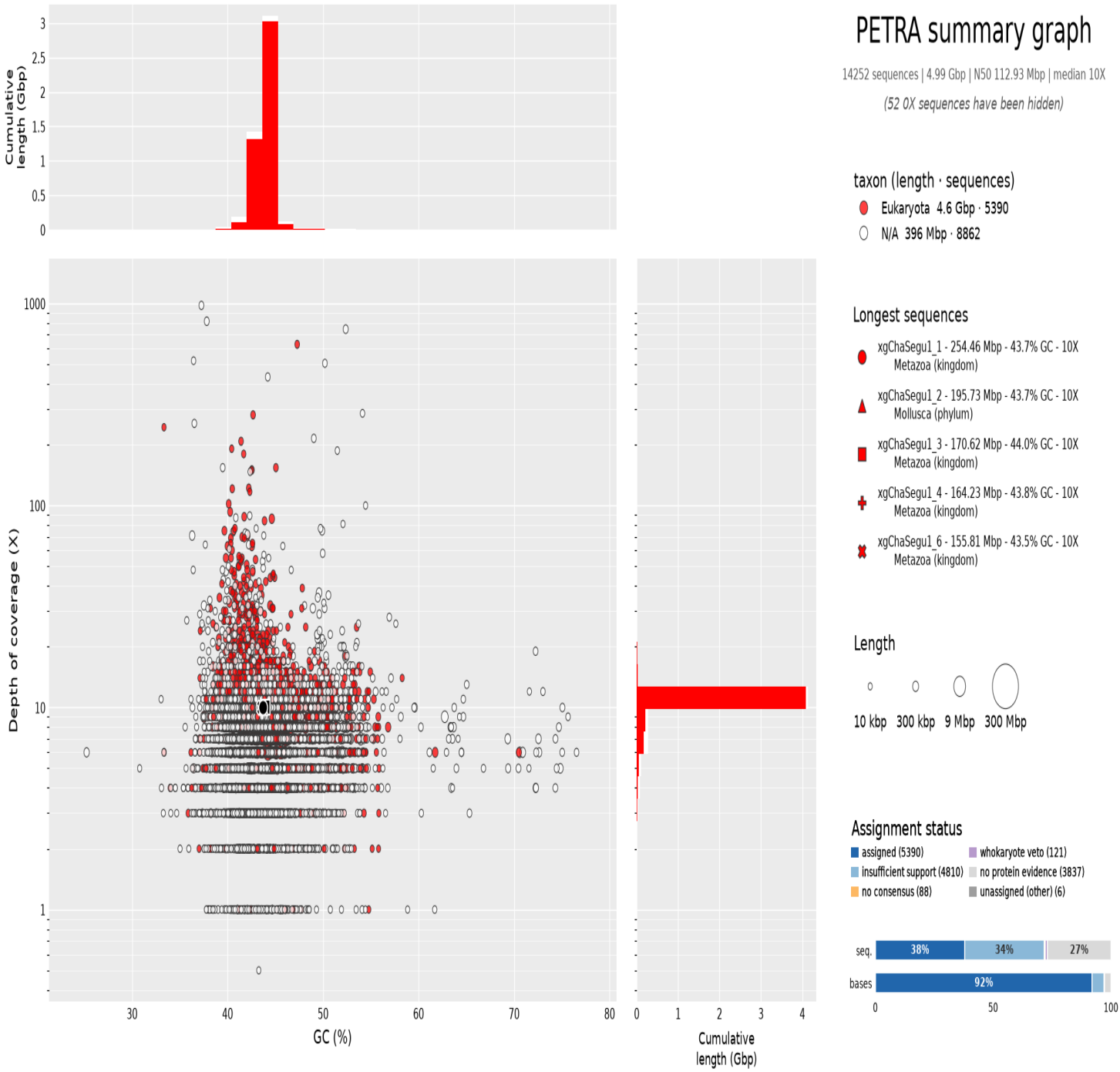


Distribution of k-mer counts per copy numbers found in asm



Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



collapsed. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	Long reads	Arima
Coverage	12	28

Assembly pipeline

- **Hifiasm**
  |_ *ver:* 0.19.5-r593
  |_ *key param:* NA
- **purge_dups**
  |_ *ver:* 1.2.5
  |_ *key param:* NA
- **YaHS**
  |_ *ver:* 1.2
  |_ *key param:* NA

Curation pipeline

- **PretextMap**
  |_ *ver:* 0.2.4
  |_ *key param:* NA
- **PretextView**
  |_ *ver:* 1.0.7
  |_ *key param:* NA

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